

Beekeeping – Basic Honor



Skill Level: 1

Category: Nature



Brief History of Beekeeping

Beekeeping has been practiced for thousands of years. Early people collected honey from wild hives. Ancient Egyptians kept bees in clay hives and used honey for food, medicine, and religious purposes. Modern beekeeping began with the invention of the movable-frame hive, which allows honey to be harvested without destroying the colony.

Uses of Bee Products

Honey

Natural sweetener

Food preservative

Medicine for coughs and wounds

Source of quick energy

Used in cooking and baking

Beeswax

Candle making

Cosmetics and lip balms

Polishes and waterproofing

Art and crafts

Lubricants

Propolis

Natural antibiotic

Seals cracks in the hive

Strengthens hive structure

Used in medicine

Protects bees from disease

Foods That Depend on Honey Bees

Many crops rely heavily on honey bee pollination:

Apples

Almonds

Cucumbers

Watermelons

Pumpkins

Avocados

Blueberries

Strawberries

Sunflowers

Peaches

Types of Bees and Their Duties

Queen Bee (Honey Bee)

Lays all eggs in the colony

Maintains colony unity through pheromones

Worker Bees (Honey Bees)

Collect nectar and pollen

Build wax comb

Feed larvae and queen

Guard and clean the hive

Drone Bees (Honey Bees)

Mate with the queen

Do not collect food or build comb

Other Types of Bees (Non-Honey Bees)

Carpenter Bees

Large bees that drill holes in wood to make nests

Important pollinators

Do not produce honey

Bumble Bees

Large, fuzzy bees

Live in small colonies

Excellent pollinators, especially in cold weather

Mason Bees

Solitary bees

Nest in holes in wood or stems

Use mud to seal nests

Leafcutter Bees

Cut pieces of leaves to build nests

Solitary bees

Very effective pollinators

Sweat Bees

Small bees attracted to human sweat

Solitary or semi-social

Important pollinators

Mining Bees

Nest underground

Solitary bees

Active early in spring

Importance of Different Bee Types

Most bees are pollinators

Many do not sting unless threatened

Not all bees make honey

All play an important role in plant reproduction

How Bees Build Comb

Worker bees secrete wax from glands on their abdomen. The wax is molded into hexagonal cells. Comb darkens with age due to pollen stains, propolis, and repeated use for raising brood.

Beekeeping Terms

Movable-frame hive: A hive designed with removable frames for inspection and honey extraction

Crossed comb: Comb built incorrectly across frames, making inspection difficult

Bee space: The proper space bees leave between combs to move freely

Swarming: When part of a colony leaves with a queen to form a new hive

Smoker

A smoker produces cool smoke that calms bees and reduces defensive behavior. Good smoker fuels include dry grass, pine needles, wood shavings, burlap, cardboard, and dry leaves.

Choosing a Hive Location

Receives morning sunlight

Sheltered from strong winds

Close to a clean water source

Near flowering plants

Safe distance from people and animals

Bee Diseases and How They Spread

Bee diseases spread through drifting bees, robbing, shared equipment, contaminated combs, and weak colonies.

Robbing

Robbing occurs when strong colonies steal honey from weaker ones. Robber bees act aggressive and fight at hive entrances.

Preventing Swarming

Provide enough space in the hive

Add supers early

Replace old or weak queens

Remove swarm cells when found

Swarming should be prevented to keep colonies productive and strong.

Winter Survival Requirements

For a colony to survive winter, it must have:

A strong, healthy queen

Enough stored honey

Protection from cold, wind, and moisture

Advantage of a Double-Brood Chamber

A double-brood chamber provides more space for brood rearing and food storage, resulting in stronger colonies and better honey production.

Duties of Successful Beekeeping

Spring feeding to stimulate brood production

Adding supers and using swarm-prevention methods

Extracting honey and storing it properly

Fall feeding and preparing the hive for winter

When Honey Is Ready for Harvest

Honey is ready for extraction when the cells are sealed (capped) with wax.

How to Make a Simple Beehive (Basic Notes)

A common beehive used today is the movable-frame hive (Langstroth hive).

Basic steps:

Build a wooden hive box using smooth, untreated wood

Ensure correct bee space between frames

Add removable frames inside the box

Place the hive on a stand above ground

Add a roof to protect from rain and sun

Hives should be strong, dry, and easy to open for inspection.

Parts of a Beehive and Their Functions

Hive Stand – Keeps the hive off the ground and protects from moisture and pests

Bottom Board – Floor of the hive; entrance for bees

Entrance – Allows bees to enter and exit

Brood Box (Brood Chamber) – Where the queen lays eggs and young bees are raised

Frames – Hold the comb; removable for inspection

Foundation – Guides bees in building straight comb

Supers (Honey Supers) – Boxes placed above the brood chamber for honey storage

Inner Cover – Provides ventilation and insulation

Outer Cover (Roof) – Protects hive from weather

How Honey Is Collected

Honey supers are removed when frames are capped

Bees are gently brushed or smoked away

Frames are taken to a clean extraction area

Wax caps are removed using a knife or uncapping tool

Frames are placed in a honey extractor and spun

How Honey Is Stored

Extracted honey is strained to remove wax and debris

Honey is poured into clean, dry containers

Stored in sealed jars

Kept in a cool, dry place away from direct sunlight

Types of Bees vs Beekeeping Bees

Types of Bees

Honey Bees – Live in colonies, produce honey and wax, used in beekeeping

Carpenter Bees – Large bees that bore into wood; do not make honey

Bumblebees – Fuzzy bees, live in small colonies, good pollinators but not kept for honey

Solitary Bees – Live alone (e.g., mason bees); no honey production

Bees Used in Beekeeping

Honey bees (*Apis mellifera*) are the main bees used in beekeeping

They produce large amounts of honey

Can be managed safely in hives

Live in organized colonies with queen, workers, and drones

Parts of a Beehive (Quick Review)

Hive Stand – Raises hive off the ground

Bottom Board – Floor of the hive

Entrance – Opening where bees enter and exit

Brood Box – Area where queen lays eggs

Frames – Hold comb and can be removed

Foundation – Helps bees build straight comb

Honey Supers – Boxes where honey is stored

Inner Cover – Helps with airflow and insulation

Outer Cover (Roof) – Protects hive from weather

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